

ADAR Platform & Architecture

Module reference -- ADAR Knowledge Academy -- Agonomia Labs Product Tour

ADAR is Agonomia Labs' shared cloud-native platform: one reusable stack of identity, storage, retrieval, and AI orchestration that every product vertical -- DocIntel, Knowledge Academy, Conversation Assistant, and the industry apps -- is built on top of. This document is the companion reading for the module's video lesson; it covers what the video shows visually, in a form the AI Tutor and Study Tools can cite directly.

System diagram

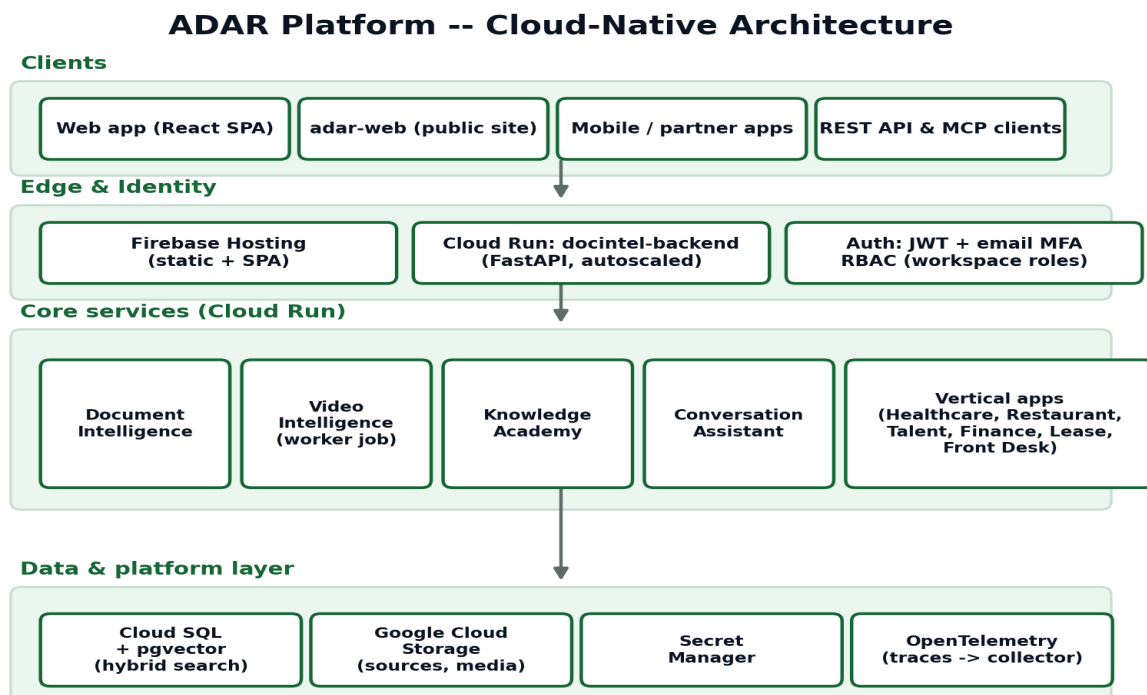


Figure 1. The four layers of the ADAR platform, from client to data store.

Layer by layer

Clients

Every surface -- the DocIntel web app, this public adar-web site, mobile and partner integrations, and direct REST/MCP clients -- talks to the same backend API. There is no separate code path for "the demo" versus "the product": what a real signed-in user sees and what this guest preview sees are served by the identical endpoints, gated by role.

Edge & identity

- **Firebase Hosting** serves the static site and the compiled React single-page app.
- **Cloud Run (docintel-backend)** runs the FastAPI application behind it: stateless, autoscaled containers with no server to patch or provision by hand.
- **Auth** issues short-lived JWTs after email/password login, with an optional email-MFA step (see the Security & Governance module), and every request is authorized against a workspace role -- owner, editor, or viewer.

Core services

Document Intelligence, Video Intelligence, Knowledge Academy, the Conversation Assistant, and each industry vertical (Healthcare, Restaurant, Talent Management, Finance & Tax, Lease Intelligence, Front Desk) run as independently deployable services on Cloud Run. Long-running work -- transcribing and indexing a video, for example -- runs on a separate Cloud Run Job so it never blocks the request/response path of the main API.

Data & platform layer

- **Cloud SQL + pgvector** stores structured records and vector embeddings side by side, so retrieval can combine keyword (BM25-style) search with semantic similarity in one query.
- **Google Cloud Storage** holds every uploaded source file and processed media asset.
- **Secret Manager** holds credentials and API keys -- never environment files or source control.
- **OpenTelemetry** instruments every service, feeding the Observability module's trace view.

Why this matters for the AI Tutor

Because every vertical shares the same retrieval and generation pipeline, a grounded answer from the AI Tutor in this Knowledge Academy demo uses the exact same hybrid-search-plus-rerank pipeline that DocIntel uses for a paying customer's private documents -- just scoped to this course's public, published content instead of a workspace's private files.